Team Notebook

[UNI-FIIS] - SaleEnOdeUno

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1 Data Structure

1.1 2d fenwick tree

```
struct BIT2D {
    vector<vector<int>> t;
    int n, m;
    BIT2D(int n, int m) : n(n), m(m) {
        t.assign(n, vector<int>(m, 0));
    }
    void update(int x, int y, int add) {
        for (int i = x; i < n; i |= i + 1)
            for (int j = y; j < m; j |= j + 1) t[i][j] +=
add;
    }
    int sum(int x, int y) {
        int ans = 0;
        for (int i = x; i >= 0; i = (i & (i + 1)) - 1)
            for (int j = y; j >= 0; j = (j & (j + 1)) - 1)
ans += t[i][j];
        return ans;
    }
    int query(int lx, int rx, int ly, int ry) {
        return sum(rx, ry) - sum(lx - 1, ry) - sum(rx, ly -
1) + sum(lx - 1, ly - 1);
    }
};
```

1.2 centroid decomposition

```
struct Centroid_Decomposition{
    vector<int> adj[N];
    int sz[N],vis[N],aux;
    int pa[N];

    int find(int u,int p,int Sz){
        int mx = 0, c = 0; sz[u] = 1;
        for (auto v : adj[u]){
            if (v == p || vis[v]) continue;
            c ^= find(v, u, Sz);
            sz[u] += sz[v];
            mx = max(mx, sz[v]);
        }
        if (max(mx, Sz-sz[u]) <= Sz/2)
            aux = p, c = u;
        return c;
    }

    void build(int u,int p,int Sz){
        aux = -1, u = find(u, -1, Sz);
        if (aux != -1) sz[aux] = Sz-sz[u];
        vis[u] = 1, sz[u] = Sz, pa[u] = p;
        for (auto v : adj[u])
            if (!vis[v]) build(v, u, sz[v]);
    }
} cd; // by NekoRolly
```

1.3 dynamic segment tree

```
struct nod {
    ll t;
    ll lazy;
    nod *l;
    nod *r;
};
static nod nods[ND];
static int id = 0;
nod *crearnodo() {
    nods[id].t = NEUT, nods[id].lazy = NEUT_LAZY, nods[id].l
= nods[id].r = NULL;
    return &nods[id++];
}
ll f(ll a, ll b) {
    return a + b;
}
void aply(ll val, nod *u, ll l, ll r) {
    u->t = (r - l + 1) * val;
    u->lazy = val;
}
struct segmentTree {
    ll n;
    nod *root = NULL;
    segmentTree(ll n) : n(n) {}
    void push(nod *u, ll l, ll r) {
        if (u->lazy == NEUT_LAZY || l == r) return;
        ll m = (l + r) / 2;
        if (!u->l) u->l = crearnodo();
        if (!u->r) u->r = crearnodo();
        aply(u->lazy, u->l, l, m);
        aply(u->lazy, u->r, m + 1, r);
        u->lazy = NEUT_LAZY;
    }
    ll query(ll ql, ll qr, nod *u, ll l, ll r) {
        if (qr < l || r < ql || !u) return NEUT;
        if (ql <= l && r <= qr) return u->t;
        push(u, l, r);
        ll m = (l + r) / 2;
        return f(query(ql, qr, u->l, l, m), query(ql, qr, u-
>r, m + 1, r));
    }
    ll query(ll ql, ll qr) {
        return query(ql, qr, root, 0, n - 1);
    }
    void update(ll ql, ll qr, ll val, nod *&u, ll l, ll r) {
        if (qr < l || r < ql) return;
        if (!u) u = crearnodo();
        if (ql <= l && r <= qr) aply(val, u, l, r);
        else {
            push(u, l, r);
            ll m = (l + r) / 2;
            update(ql, qr, val, u->l, l, m);
            update(ql, qr, val, u->r, m + 1, r);
            u->t = f(u->l ? u->l->t : NEUT, u->r ? u->r->t :
NEUT);
        }
    }
    void update(ll ql, ll qr, ll val) {
        update(ql, qr, val, root, 0, n - 1);
    }
};
```

1.4 fenwick tree

```
struct FenwickTree { // Index 1
    int n, ft[MX + 1];
    void update(int l, int v) {
        for (; l <= n; l += l & -l) ft[l] += val;
    }
    int query(int l) {
        int ans = 0;
        for (; l > 0; l -= l & -l) ans += t[l];
        return ans;
    }
    int query(int l, int r) { return query(r) - query(l -
1); }
}
```

1.5 hld

```
struct Graph {
    vector<vector<ll>> adj;
    vector<ll> padre, prof, heavy, head, pos, base, L, R;
    segmentTree st;
    ll curpos, n;
    ll dfs(ll u, ll p) {
        padre[u] = p;
        ll szu = 1;
        ll mxsz = 0;
        heavy[u] = 0;
        for (ll v : adj[u])
            if (v != p) {
                prof[v] = prof[u] + 1;
                ll szv = dfs(v, u);
                szu += szv;
                if (mxsz < szv) {
                    mxsz = szv;
                    heavy[u] = v;
                }
            }
        return szu;
    }
    void decompose(ll u, ll h, vector<ll> &a) {
        head[u] = h;
        L[u] = curpos;
```

```
        pos[u] = curpos++;
        base[pos[u]] = a[u - 1];
        if (heavy[u] != 0) decompose(heavy[u], h, a);
        for (ll v : adj[u])
            if (v != padre[u] && v != heavy[u]) decompose(v,
v, a);
        R[u] = curpos - 1;
    }
    void pupdate(ll u, ll val) {
        st.pupdate(pos[u], val);
    }
    void update(ll u, ll v, ll val) {
        while (head[u] != head[v]) {
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
            st.update(pos[head[v]], pos[v], val);
            v = padre[head[v]];
        }
        if (prof[u] > prof[v]) swap(u, v);
        st.update(pos[u], pos[v], val);
    }
    ll query(ll u, ll v) {
        ll ans = INF;
        while (head[u] != head[v]) {
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
            ans = f(ans, st.query(pos[head[v]], pos[v]));
            v = padre[head[v]];
        }
        if (prof[u] > prof[v]) swap(u, v);
        ans = f(ans, st.query(pos[u], pos[v]));
        return ans;
    }
    ll subtreequery(ll u) {
        return st.query(L[u], R[u]);
    }
    void subtreeupdate(ll u, ll val) {
        st.update(L[u], R[u], val);
    }
    Graph(vector<ll> &a, vector<pll> &aristas, ll root =
1) : n(a.size()) {
        adj.assign(n + 1, vector<ll>()), padre.assign(n + 1,
0), prof.assign(n + 1, 0);
        heavy.assign(n + 1, 0), head.assign(n + 1, 0),
pos.assign(n + 1, 0);
        L.assign(n + 1, 0), R.assign(n + 1, 0),
base.resize(n);
        curpos = 0;
        for (auto [u, v] : aristas) {
            adj[u].pb(v);
            adj[v].pb(u);
        }
        prof[root] = 0;
        dfs(root, root);
        decompose(root, root, a);
        st = segmentTree(base);
    }
};
```

1.6 hld edge

```
struct Graph {
    vector<vector<pll>> adj;
    vector<ll> padre, wpadre, prof, heavy, head, pos, base,
L, R;
    segmentTree st;
    ll curpos;
    ll n;
    ll dfs(ll u, ll p, ll wp) {
        padre[u] = p;
        wpadre[u] = wp;
        ll szu = 1;
        ll mxsz = 0;
        heavy[u] = 0;
        for (auto [v, w] : adj[u])
            if (v != p) {
                prof[v] = prof[u] + 1;
                ll szv = dfs(v, u, w);
                szu += szv;
                if (mxsz < szv) mxsz = szv, heavy[u] = v;
            }
        return szu;
    }
    void decompose(ll u, ll h) {
        head[u] = h;
        L[u] = curpos;
        pos[u] = curpos++;
        base[pos[u]] = wpadre[u];
        if (heavy[u] != 0) decompose(heavy[u], h);
        for (auto [v, w] : adj[u])
            if (v != padre[u] && v != heavy[u]) decompose(v,
v);
        R[u] = curpos - 1;
    }
    void pupdate(ll u, ll v, ll val) {
        if (u == padre[v]) swap(u, v);
        st.pupdate(pos[u], val);
    }
    void update(ll u, ll v, ll val) {
        while (head[u] != head[v]) {
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
            st.update(pos[head[v]], pos[v], val);
            v = padre[head[v]];
        }
        if (prof[u] > prof[v]) swap(u, v);
        if (u != v) st.update(pos[u] + 1, pos[v], val);
    }
    ll query(ll u, ll v) {
        ll ans = INF;
        while (head[u] != head[v]) {
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
            ans = f(ans, st.query(pos[head[v]], pos[v]));
            v = padre[head[v]];
        }
        if (prof[u] > prof[v]) swap(u, v);
        if (u != v) ans = f(ans, st.query(pos[u] + 1,
pos[v]));
        return ans;
    }
    ll subtreequery(ll u) {
        if (L[u] == R[u]) return 0;
        return st.query(L[u] + 1, R[u]);
    }
    void subtreeupdate(ll u, ll val) {
        if (L[u] == R[u]) return;
        st.update(L[u] + 1, R[u], val);
    }
    Graph(vector<pair<pll, ll>> &aristas, ll root = 1) :
n(aristas.size() + 1) {
        adj.assign(n + 1, vector<pll>());
        padre.assign(n + 1, 0), wpadre.assign(n + 1, 0),
prof.assign(n + 1, 0);
        heavy.assign(n + 1, 0), head.assign(n + 1, 0),
pos.assign(n + 1, 0);
        L.assign(n + 1, 0), R.assign(n + 1, 0),
base.resize(n);
        curpos = 0;
        for (auto [edge, w] : aristas) {
            auto [u, v] = edge;
            adj[u].pb({v, w});
            adj[v].pb({u, w});
        }
        prof[root] = 0;
        dfs(root, root, 0LL);
        decompose(root, root);
        st = segmentTree(base);
    }
};
```

1.7 merge sort tree

```
struct segmentTree {
    vector<vector<ll>> seg;
    ll n, N;
    segmentTree() {}
    segmentTree(vector<ll> &a) : n(a.size()) {
        N = 1;
        while (N < n) N <<= 1;
        seg.assign(2 * N, vector<ll>());
        for (int i = 0; i < n; i++) seg[i + N] = {a[i]};
        for (int i = N - 1; i > 0; i--) {
            auto &l = seg[i << 1], &r = seg[i << 1 | 1];
            seg[i].resize((ll)l.size() + (ll)r.size());
            merge(all(l), all(r), seg[i].begin());
        }
    }
```

```
    // cantidad de elementos mayores a "a" en el intervalo
[ql,qr]
    ll query1(ll ql, ll qr, ll a) {
        ll l = ql + N, r = qr + N;
        ll ans = 0;
        while (l <= r) {
            if (l & 1) {
                auto &v = seg[l++];
                ans += v.end() - upper_bound(all(v), a);
            }
            if (!(r & 1)) {
                auto &v = seg[r--];
                ans += v.end() - upper_bound(all(v), a);
            }
            l >>= 1;
            r >>= 1;
        }
        return ans;
    }
    // cantidad de elementos menores a "a" en el intervalo
[ql,qr]
    ll query2(ll ql, ll qr, ll a) {
        ll l = ql + N, r = qr + N;
        ll ans = 0;
        while (l <= r) {
            if (l & 1) {
                auto &v = seg[l++];
                ans += lower_bound(all(v), a) - v.begin();
            }
            if (!(r & 1)) {
                auto &v = seg[r--];
                ans += lower_bound(all(v), a) - v.begin();
            }
            l >>= 1;
            r >>= 1;
        }
        return ans;
    }
};
```

1.8 multi ordered set

```
// find_by_order(k) -> return iterator to the k-th element
(0-indexed) - O(log n)
// order_of_key(num) -> # of items strictly smaller than num
- O(log n)

// using less_equal cause that lower_bound works as
upper_bound and viceversa
// also .find() does not work and you can only erase by
iterator now
// using for example erase(upper_bound(x))
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;

template <typename T>
using ordered_set =
    tree<T, null_type, less_equal<T>, rb_tree_tag,
tree_order_statistics_node_update>;

ordered_set<int> s;
```

1.9 ordered set

```
// find_by_order(k) -> return iterator to the k-th element
(0-indexed) - O(log n)
// order_of_key(num) -> # of items strictly smaller than num
- O(log n)
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;

template <typename T>
using ordered_set =
    tree<T, null_type, less<T>, rb_tree_tag,
tree_order_statistics_node_update>;

ordered_set<int> S;
```

1.10 persistent segment tree

```
const ll MX = 2e5 + 5;
const ll Q = 2e5 + 5;
const ll MXT = 2 * MX + 20 * Q;
struct PST {
    ll root[Q + 1], t[MXT], L[MXT], R[MXT];
    ll n, nodos, lastversion = 0;
    PST(vector<ll> &a) : n(a.size()) {
        nodos = 2 * n - 1;
        root[0] = 1;
        build(a, 1, 0, n - 1);
    }
    ll f(ll a, ll b) {
        return a + b;
    }
    void build(vector<ll> &a, ll id, ll l, ll r) {
        if (l == r) t[id] = a[l];
        else {
            ll m = (l + r) / 2;
            ll idl = id + 1;
            ll idr = id + 2 * (m - l + 1);
            L[id] = idl;
            R[id] = idr;
            build(a, idl, l, m);
            build(a, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);
        }
    }
    ll query(ll ql, ll qr, ll id, ll l, ll r) {
        if (ql <= l && r <= qr) return t[id];
        ll m = (l + r) / 2;
        if (qr <= m) return query(ql, qr, L[id], l, m);
        if (m + 1 <= ql) return query(ql, qr, R[id], m + 1,
r);
        return f(query(ql, qr, L[id], l, m), query(ql, qr,
R[id], m + 1, r));
    }
    ll query(ll ql, ll qr, ll version = -1) {
        if (version == -1) version = lastversion;
        return query(ql, qr, root[version], 0, n - 1);
    }
    void update(ll pos, ll val, ll prev, ll &id, ll l, ll r)
{
        id = ++nodos;
        if (l == r) t[id] = val;
        else {
            ll m = (l + r) / 2;
            L[id] = L[prev];
            R[id] = R[prev];
            if (pos <= m) update(pos, val, L[prev], L[id],
l, m);
            else update(pos, val, R[prev], R[id], m + 1, r);
            t[id] = f(t[L[id]], t[R[id]]);
        }
    }
    void update(ll pos, ll val, ll version = -1) {
        if (version == -1) version = lastversion;
        lastversion++;
        update(pos, val, root[version], root[lastversion],
0, n - 1);
    }
};
```

1.11 persistent segment tree lazy

```
const int MX = 2e5 + 5;
const int Q = 2e5 + 5;
const int LOG = 20;
const int ND = 2 * MX + 6 * Q * LOG;
const ll NEUT = 0;
const ll NEUT_LAZY = -2 * INF;
struct nod {
    ll t, lazy;
    nod *l, *r;
};
nod nods[ND];
int id = 0;
nod *crearnodo(nod *u = NULL) {
    if (u) nods[id].t = u->t, nods[id].lazy = u->lazy,
nods[id].l = u->l, nods[id].r = u->r;
    else nods[id].t = NEUT, nods[id].lazy = NEUT_LAZY,
nods[id].l = nods[id].r = NULL;
    return &nods[id++];
}
ll f(ll a, ll b) {
    return a + b;
```

```
}
void apply(ll val, nod *u, ll l, ll r) {
    u->t += (r - l + 1) * val;
    if (u->lazy == NEUT_LAZY) u->lazy = val;
    else u->lazy += val;
}
struct PST {
    nod *root[MX + 5];
    ll n;
    ll lastversion = -1;
    PST(vector<ll> &a) : n(a.size()) {
        root[++lastversion] = crearnodo();
        build(a, root[0], 0, n - 1);
    }
    void build(vector<ll> &a, nod *&u, ll l, ll r) {
        if (!u) u = crearnodo();
        if (l == r) u->t = a[l];
        else {
            ll m = (l + r) >> 1;
            build(a, u->l, l, m);
            build(a, u->r, m + 1, r);
            u->t = f(u->l->t, u->r->t);
        }
    }
    void push(nod *u, ll l, ll r) {
        if (u->lazy == NEUT_LAZY || l == r) return;
        ll m = (l + r) >> 1;
        u->l = crearnodo(u->l), u->r = crearnodo(u->r);
        apply(u->lazy, u->l, l, m);
        apply(u->lazy, u->r, m + 1, r);
        u->lazy = NEUT_LAZY;
    }
    nod *update(ll ql, ll qr, ll val, nod *prev, ll l, ll r)
{
        if (qr < l || r < ql) return prev;
        nod *u = crearnodo(prev);
        if (ql <= l && r <= qr) apply(val, u, l, r);
        else {
            push(u, l, r);
            ll m = (l + r) >> 1;
            u->l = update(ql, qr, val, u->l, l, m);
            u->r = update(ql, qr, val, u->r, m + 1, r);
            u->t = f(u->l->t, u->r->t);
        }
        return u;
    }
    void update(ll ql, ll qr, ll val, ll version = -1) {
        if (version == -1) version = lastversion;
        root[++lastversion] = update(ql, qr, val,
root[version], 0, n - 1);
    }
    ll query(ll ql, ll qr, nod *u, ll l, ll r) {
        if (!u || qr < l || r < ql) return NEUT;
        if (ql <= l && r <= qr) return u->t;
        push(u, l, r);
        ll m = (l + r) >> 1;
        return f(query(ql, qr, u->l, l, m), query(ql, qr, u-
>r, m + 1, r));
    }
    ll query(ll ql, ll qr, ll version = -1) {
        if (version == -1) version = lastversion;
        return query(ql, qr, root[version], 0, n - 1);
    }
};
```

1.12 segment tree

```
struct IterativeSegmentTree { // Index 0
    int n;
    ll ST[2 * MX];
    const ll NEUT = 0;
    void build(vector<ll> &a) {
        n = a.size();
        for (int i = n; i < (n << 1); i++) ST[i] = a[i - n];
        for (int i = n - 1; i > 0; i--) ST[i] = f(ST[i <<
1], ST[i << 1 | 1]);
    }
    ll f(ll x, ll y) { return max(x, y); }
    void update(int i, ll val) {
        for (ST[i += n] = val; i >>= 1; ) ST[i] = f(ST[i <<
1], ST[i << 1 | 1]);
    }
    ll query(int l, int r) { // [l, r>
        ll ansL = NEUT, ansR = NEUT;
        for (l += n, r += n; l < r; l >>= 1, r >>= 1) {
            if (l & 1) ansL = f(ansL, ST[l++]);
            if (r & 1) ansR = f(ST[--r], ansR);
        }
        return f(ansL, ansR);
    }
};
```

1.13 segment tree 2d

```
struct ST2D { //Index 0
    int n, m;
    vector<vector<ll>> t;
    ll f(ll a, ll b) {
        return a + b;
    }
    ST2D(vector<vector<ll>> &a) {
        n = a.size();
        m = a[0].size();
        t.assign(2 * n, vector<ll>(2 * m, 0));
        for (int i = 0; i < n; i++)
            for (int j = 0; j < m; j++) t[i + n][j + m] =
a[i][j];
        for (int i = n; i < 2 * n; i++)
            for (int j = m - 1; j > 0; j--) t[i][j] = f(t[i]
[j << 1], t[i][j << 1 | 1]);
        for (int i = n - 1; i > 0; i--)
            for (int j = 1; j < 2 * m; j++) t[i][j] = f(t[i
<< 1][j], t[i << 1 | 1][j]);
    }
    void update(int x, int y, ll val) {
        x += n, y += m, t[x][y] = val;
        for (int j = y; j > 1; j >>= 1) t[x][j >> 1] =
f(t[x][j], t[x][j ^ 1]);
        for (int i = x >> 1; i > 0; i >>= 1) {
            int j = y;
            t[i][j] = f(t[i << 1][j], t[i << 1 | 1][j]);
            for (; j > 1; j >>= 1) t[i][j >> 1] = f(t[i][j],
t[i][j ^ 1]);
        }
    }
    ll queryY(int i, int l, int r) {
        ll ans = 0;
        for (l += m, r += m + 1; l < r; l >>= 1, r >>= 1) {
            if (l & 1) ans = f(ans, t[i][l++]);
            if (r & 1) ans = f(t[i][--r], ans);
        }
        return ans;
    }
    ll query(int lx, int rx, int ly, int ry) {
        ll ans = 0;
        for (x1 += n, x2 += n + 1; x1 < x2; x1 >>= 1, x2 >>=
1) {
            if (x1 & 1) ans = f(ans, queryY(x1++, y1, y2));
            if (x2 & 1) ans = f(queryY(--x2, y1, y2), ans);
        }
        return ans;
    }
};
```

1.14 segment tree beats

```
struct node {
    ll sum;
    ll max1, max2, cont; // min para chmax
};
node f(node a, node b) { // simetrico con min para chmax
    node ans;
    ans.sum = a.sum + b.sum;
    ans.max1 = max(a.max1, b.max1);
    if (a.max1 == b.max1) ans.max2 = max(a.max2, b.max2),
ans.cont = a.cont + b.cont;
    else if (a.max1 > b.max1) ans.max2 = max(a.max2,
b.max1), ans.cont = a.cont;
    else ans.max2 = max(a.max1, b.max2), ans.cont = b.cont;
    return ans;
}
struct SegmentTree {
    vector<node> t;
    int n;
    SegmentTree(vector<ll> &a) : n(a.size()) {
```

```
        t.resize(2 * n);
        build(a, 1, 0, n - 1);
    }
    void build(vector<ll> &a, int id, int l, int r) {
        if (l == r) t[id] = {a[l], a[l], -INF, 1}; // chmax:
-INF -> INF
        else {
            int m = (l + r) >> 1;
            int idl = id + 1;
            int idr = id + 2 * (m - l + 1);
            build(a, idl, l, m), build(a, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);
        }
    }
    void apply(ll val, int id) {
        if (t[id].max1 <= val) return; // chmax: (<=) ->
(>=)
        t[id].sum += t[id].cont * (val - t[id].max1);
        t[id].max1 = val;
    }
    void push(int id, int l, int r) {
        if (l == r) return;
        int m = (l + r) >> 1;
        int idl = id + 1;
        int idr = id + 2 * (m - l + 1);
        apply(t[id].max1, idl);
        apply(t[id].max1, idr);
    }
    void update(int ql, int qr, ll val, int id, int l, int
r) {
        if (qr < l || r < ql || t[id].max1 <= val)
return; //chmax: (<=) -> (>=)
        if (ql <= l && r <= qr && t[id].max2 < val)
apply(val, id); // chmax: (<) -> (>)
        else {
            push(id, l, r);
            int m = (l + r) >> 1;
            int idl = id + 1;
            int idr = id + 2 * (m - l + 1);
            update(ql, qr, val, idl, l, m), update(ql, qr,
val, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);
        }
    }
    ll query(int ql, int qr, int id, int l, int r) {
        if (qr < l || r < ql) return NEUT;
        if (ql <= l && r <= qr) return t[id].sum;
        push(id, l, r);
        int m = (l + r) >> 1;
        int idl = id + 1;
        int idr = id + 2 * (m - l + 1);
        return query(ql, qr, idl, l, m) + query(ql, qr, idr,
m + 1, r);
    }
};
```

1.15 segment tree lazy

```
struct SegmentTree { // id: index 1. Vector a: Index 0
    vector<ll> t, lazy;
    int n;
    SegmentTree(vector<ll> &a) : n(a.size()) {
        t.assign(2 * n, NEUT), lazy.assign(2 * n,
NEUT_LAZY);
        build(a, 1, 0, n - 1);
    }
    ll f(ll a, ll b) {
        return a + b;
    }
    void aply(ll val, int id, int l, int r) {
        t[id] += (r - l + 1) * val;
        lazy[id] += val;
    }
    void build(vector<ll> &a, int id, int l, int r) {
        if (l == r) t[id] = a[l];
        else {
            int m = (l + r) >> 1;
            int idl = id + 1;
            int idr = id + 2 * (m - l + 1);
            build(a, idl, l, m), build(a, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);
        }
    }
    void push(int id, int l, int r) {
        if (l == r || lazy[id] == NEUT_LAZY) return;
        int m = (l + r) >> 1;
        int idl = id + 1;
        int idr = id + 2 * (m - l + 1);
        aply(lazy[id], idl, l, m);
        aply(lazy[id], idr, m + 1, r);
        lazy[id] = NEUT_LAZY;
    }
    void update(int ql, int qr, ll val, int id, int l, int
r) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) aply(val, id, l, r);
        else {
            push(id, l, r);
            int m = (l + r) >> 1;
            int idl = id + 1;
            int idr = id + 2 * (m - l + 1);
            update(ql, qr, val, idl, l, m), update(ql, qr,
val, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);
        }
    }
    ll query(int ql, int qr, int id, int l, int r) {
        if (qr < l || r < ql) return NEUT;
        if (ql <= l && r <= qr) return t[id];
        push(id, l, r);
        int m = (l + r) >> 1;
        int idl = id + 1;
        int idr = id + 2 * (m - l + 1);
        return f(query(ql, qr, idl, l, m), query(ql, qr,
idr, m + 1, r));
    }
};
```

1.16 sparse table

```
const int LG = __lg(MX) + 1;
int st[MX][LG];
void build(vector<int> &A) { // O(|f| n log n)
    int n = A.size();
    for (int i = 0; i < n; i++) st[i][0] = A[i];
    for (int k = 1; k < LG; k++) {
        for (int i = 0; i + (1 << k) <= n; i++) {
            st[i][k] = min(st[i][k - 1], st[i + (1 << (k -
1))][k - 1]);
        }
    }
}
int query(int l, int r) { // O(|f|)
    int lg = __lg(r - l + 1);
    return min(st[l][lg], st[r - (1 << lg) + 1][lg]);
}
```

1.17 trie bitmask

```
struct Trie{
    int t[N*B][2],sz[N*B];
    int cur;

    void insert(int x){
        int u = 0; sz[u]++;
        for (int i=B-1; i>=0; i--){
            int k = x>>i&1;
            if (t[u][k] == 0) t[u][k] = ++cur;
            u = t[u][k], sz[u]++;
        }
    }

    void erase(int x){
        int u = 0; sz[0]--;
        for (int i=B-1; i>=0; i--){
            int k = x>>i&1;
            u = t[u][k], sz[u]--;
        }
    }

    int get_min(int x){ // min xor
        int u = 0, ans = 0;
        for (int i=B-1; i>=0; i--){
            int k = x>>i&1;
            if (t[u][k] != 0 && sz[t[u][k]] > 0) u = t[u]
[k];
```

```
            else u = t[u][k^1], ans ^= 1<<i;
        }
        return ans;
    }

    int get_max(int x){ // max xor
        int u = 0, ans = 0;
        for (int i=B-1; i>=0; i--){
            int k = x>>i&1;
            if (t[u][k^1] != 0 && sz[t[u][k^1]] > 0)
                u = t[u][k^1], ans ^= 1<<i;
            else u = t[u][k];
        }
        return ans;
    }
} tr; // by NekoRolly
```

1.18 trie string

```
struct Trie{
    int t[N][K],sz[N],cur;

    void insert(string &s){
        int u = 0; sz[u]++;
        for (char &c : s){
            int k = c-'a';
            if (t[u][k] == 0) t[u][k] = ++cur;
            u = t[u][k], sz[u]++;
        }
    }

    void erase(string &s){
        int u = 0; sz[u]--;
        for (char &c : s){
            int k = c-'a';
            u = t[u][k], sz[u]--;
        }
    }
} tr; // by NekoRolly
```

2 Geometry

2.1 convex hull trick

```
struct Line{
    ll m,b;

    ll y(ll x){ return m*x+b;}
};

struct Convex_Hull_Trick{
    vector<Line> vec;
    int l = 0;

    bool check(int i){
        double x1 = double(vec[i-1].b - vec[i+1].b)/
(vec[i+1].m - vec[i-1].m);
        double x2 = double(vec[i].b - vec[i+1].b)/
(vec[i+1].m - vec[i].m);
        return x1 >= x2;
    }

    void add(Line L){
        vec.push_back(L);
        while (vec.size() >= 3 && check(vec.size()-2))
            vec.erase(vec.end()-2);
        l = min(l, int(vec.size()-1));
    }

    ll query(ll x){ // O(1) amortizado
        while (l+1 < vec.size() && vec[l].y(x) >
vec[l+1].y(x)) l++;
        return vec[l].y(x);
    }

    ll find(ll x){ // O(logn)
        int l = 0, r = vec.size();
        while (l+1 < r){
            int m = (l+r)>>1;
            if (vec[m-1].y(x) > vec[m].y(x)) l = m;
            else r = m;
        }
        return vec[l].y(x);
    }
} cht; // by NekoRolly
```

2.2 li chao tree

```
struct Line{
    ll m,b;

    ll y(ll x){ return m*x+b;}
};

struct Li_Chao_Tree{
    Line t[N*4];

    void update(int id,int l,int r,Line L){
        if (l+1 == r || t[id].b == -infll){
            if (L.y(l) > t[id].y(l)) t[id] = L;
            return;
        }
        int m = (l+r)>>1;
        if (t[id].y(m) < L.y(m)) swap(t[id], L);
        if (t[id].m > L.m) update(id<<1, l, m, L);
        else update(id<<1|1, m, r, L);
    }

    ll query(int id,int l,int r,ll x){
        if (l+1 == r) return t[id].y(x);
        int m = (l+r)>>1;
        if (x < m) return max(t[id].y(x), query(id<<1, l, m,
x));
        else return max(t[id].y(x), query(id<<1|1, m, r,
x));
    }
} lct; // by NekoRolly
```

3 Graphs

3.1 2sat

```
struct SAT { // Index_1
    vector<vector<int>> adj, adjt;
    vector<bool> used;
    vector<int> comp;
    int n;
    stack<int> st;
    SAT(int n) : n(n) {
        adj.resize(2 * n + 1), adjt.resize(2 * n + 1);
        used.assign(2 * n + 1, 0), comp.assign(2 * n + 1,
0);
    }
    void add(int p, bool bp, int q, bool bq) { // add p -> q
        p += (!bp) * n;
        q += (!bq) * n;
        adj[p].pb(q);
        adjt[q].pb(p);
    }
    void addor(int p, bool bp, int q, bool bq) { // add p or
q
        add(p, !bp, q, bq);
        add(q, !bq, p, bp);
    }
    void addxor(int p, bool bp, int q, bool bq) { // add p
xor q
        addor(p, bp, q, bq);
        addor(p, !bp, q, !bq);
    }
    void addnxor(int p, bool bp, int q, bool bq) { // add
~(p xor q)
        addor(p, !bp, q, bq);
        addor(p, bp, q, !bq);
    }
    void dfs(int u) {
        used[u] = 1;
        for (int v : adj[u])
            if (!used[v]) dfs(v);
        st.push(u);
    }
    void _dfs(int u, int c) {
        comp[u] = c;
        for (int v : adjt[u])
```

```
            if (!comp[v]) _dfs(v, c);
    }
    bool solve(vector<bool> &ans) {
        ans.resize(n + 1);
        for (int u = 1; u <= 2 * n; u++)
            if (!used[u]) dfs(u);
        int c = 0;
        while (!st.empty()) {
            int u = st.top();
            st.pop();
            if (!comp[u]) _dfs(u, ++c);
        }
        for (int p = 1; p <= n; p++) {
            if (comp[p] == comp[p + n]) return false;
            ans[p] = comp[p] > comp[p + n];
        }
        return true;
    }
};
```

3.2 bellman ford

```
// find distance from source node to all nodes.
// supports negative edge weights.
// returns true if a negative cycle is detected.
bool bellmanFord(int s, int n) { // O(VE)
    d.assign(n + 1, INF);
    vector<bool> cycle(n, false);
    bool f = false;
    for (int i = 0; i < n; i++) {
        f = false;
        for (int u = 0; u < n; u++) {
            for (auto [v, w] : adj[u]) {
                if (d[u] != INF && d[u] + w < d[v]) {
                    d[v] = d[u] + w;
                    parent[v] = u;
                    f = true;
                    if (i == n - 1) cycle[v] = true;
                }
            }
        }
        if (!f) return false;
    }
    return true;
}
```

3.3 block cut tree

```
vector<int> adj[MX], bcc[MX], bct[2 * MX], st;
// bcc[u] = componentes a la que pertenece el nodo u
// bct[u] = adj del BCT
int tin[MX], low[MX], timer;
// low[u] : minimo tin de los backedges entre todos los de
los desendientes de u en el __dfs
int comp[MX]; // representante de u en el BCT
int c;
void __dfs(int u, int p = -1) {
    tin[u] = low[u] = ++timer;
    st.pb(u);
    bool skip = 1; // Si ha  multiedges
    for (int v : adj[u]) {
        if (v == p && skip) {
            skip = 0;
            continue;
        }
        if (tin[v]) low[u] = min(low[u], tin[v]);
        else {
            __dfs(v, u);
            low[u] = min(low[u], low[v]);
            if (tin[u] <= low[v]) {
                c++;
                int x = -1;
                while (x != v) x = st.back(), st.pop_back(),
bcc[x].pb(c);
                bcc[u].pb(c);
            }
        }
    }
}
void buildBCT(int n) {
    for (int i = 1; i <= n; i++)
        if (!tin[i]) {
            __dfs(i);
            st.pop_back();
            if (bcc[i].empty()) bcc[i].pb(++c);
        }
    for (int u = 1; u <= n; u++) {
        if (bcc[u].size() == 1) comp[u] = bcc[u].back();
        else {
            comp[u] = ++c;
            for (int v : bcc[u]) bct[comp[u]].pb(v),
bct[v].pb(comp[u]);
        }
    }
}
```

3.4 blossom

```
struct Graph {
    vector<int> adj[MX];
    int p[MX];
    int match[MX];
    // match[u] is the node match with u
    // or -1 if it does not belong to the matching

    int base[MX]; // base of the flower
    bool used[MX];
    bool blossom[MX];

    void addEdge(int u, int v);
    int lca(int u, int v, int n) { // O(n)
        vector<bool> marked(n);
        while (true) {
            u = base[u]; // blossom compression
            marked[u] = true;
            if (match[u] == -1) break; // we reach the
root : unexposed vertex
            u = p[match[u]];
        }
        while (true) {
            v = base[v];
            if (marked[v]) return v;
            v = p[match[v]];
        }
        return -1;
    }

    void markPath(int u, int b, int children) {
        while (base[u] != b) {
            blossom[base[u]] = blossom[base[match[u]]] =
true;
            p[u] = children, children = match[u], u =
p[match[u]];
        }
    }

    void compress(int u, int v, int n, queue<int> &q) {
        int curBase = lca(u, v, n); // base of the flower
        assert(curBase != -1);
        fill(blossom, blossom + n, false);

        markPath(u, curBase, v);
        markPath(v, curBase, u);
        for (int i = 0; i < n; i++) {
            if (blossom[base[i]]) {
                base[i] = curBase;
                if (!used[i]) used[i] = true, q.push(i);
            }
        }
    }

    int findPath(int root, int n) {
        for (int i = 0; i < n; i++) base[i] = i, used[i] =
false, p[i] = -1;
        used[root] = true;
        queue<int> q;
        q.push(root);
        while (!q.empty()) {
            int u = q.front();
            q.pop();
            for (int to : adj[u]) {
                if (base[u] == base[to] || match[u] == to)
continue;
                if (to == root || (match[to] != -1 &&
```

```
p[match[to]] != -1))
                    compress(u, to, n, q); // we find a
blossom
                else if (p[to] == -1) {
                    // augmenting path or continue bfs
                    p[to] = u;
                    if (match[to] == -1) return to; // we
find an augmenting path
                    // we continue the bfs
                    to = match[to], used[to] = true,
q.push(to);
                }
            }
        }
        return -1;
    }

    int maxMatching(int n) { // O(n^3)
        fill(match, match + n, -1);
        int ans = 0;
        for (int u = 0; u < n; u++) {
            if (match[u] == -1) {
                int v = findPath(u, n);
                if (v != -1) {
                    ans++;
                    while (v != -1) {
                        int nextVertex = match[p[v]];
                        match[v] = p[v], match[p[v]] = v, v
= nextVertex;
                    }
                }
            }
        }
        return ans;
    }
} G;
```

3.5 dinic

```
// This implementation is for small graphs G(V, E) (|V| <=
3000 aprox)
// where O(|V|^2) memory will fit the memory limit.
using Cap = long long;
const int MX = 3000;
const Cap INF = 1e18;
struct Graph {
    vector<int> adj[MX];
    Cap cap[MX][MX];
    Cap flow[MX][MX];
    bool added[MX][MX];
    int level[MX];
    int nextEdge[MX];

    void clear(int n) {
        for (int i = 0; i < n; i++) {
            adj[i].clear();
            for (int j = 0; j < n; j++) cap[i][j] = 0,
flow[i][j] = 0, added[i][j] = false;
        }
    }

    void addEdge(int u, int v, Cap w, bool dir) {
        if (!added[u][v]) {
            adj[u].push_back(v);
            adj[v].push_back(u);
            added[u][v] = added[v][u] = true;
        }
        cap[u][v] += w;
        if (!dir) cap[v][u] += w;
    }

    Cap dfs(int u, int t, Cap f) { // O(E)
        if (u == t) return f;
        for (int &i = nextEdge[u]; i < adj[u].size(); i++) {
            int v = adj[u][i];
            Cap cf = cap[u][v] - flow[u][v];
            if (cf == 0 || level[v] != level[u] + 1)
continue;
            Cap ret = dfs(v, t, min(f, cf));
            if (ret > 0) {
                flow[u][v] += ret;
                flow[v][u] -= ret;
                return ret;
            }
        }
        return 0;
    }

    bool bfs(int s, int t, int n) { // O(E)
        fill(level, level + n, -1);
        queue<int> q({s});
        level[s] = 0;
        while (!q.empty()) {
            int u = q.front();
            nextEdge[u] = 0;
            q.pop();
            for (int v : adj[u]) {
                Cap cf = cap[u][v] - flow[u][v];
                if (level[v] == -1 && cf > 0) {
                    level[v] = level[u] + 1;
                    q.push(v);
                }
            }
        }
        return level[t] != -1;
    }

    Cap maxFlow(int s, int t, int n) {
        // General: O(E * V^2), O(E * V + V * |F|)
        // Unit Cap: O(E * min(E^(1/2) , V^(2/3)))
        // Unit Network: O(E * V^(1/2))
        // In unit network, all the edges have unit capacity
        // and for any vertex except s and t either the
        // incoming or outgoing edge is unique.
        Cap ans = 0;
        while (bfs(s, t, n)) { // O(V) iterations
            // after bfs, the graph is a DAG
            while (Cap inc = dfs(s, t, INF)) ans += inc;
        }
        return ans;
    }
} G;
```

3.6 dominator tree

```
//idom[i]=parent of i in dominator tree with root=rt, or -1
if not exists
int
n,rnk[MAXN],pre[MAXN],anc[MAXN],idom[MAXN],semi[MAXN],low[MAXN];
vector<int> g[MAXN],rev[MAXN],dom[MAXN],ord;
void dfspre(int pos){
    rnk[pos]=SZ(ord); ord.pb(pos);
    for(auto x:g[pos]){
        rev[x].pb(pos);
        if(rnk[x]==n) pre[x]=pos,dfspre(x);
    }
}
int eval(int v){
    if(anc[v]<n&&anc[anc[v]]<n){
        int x=eval(anc[v]);
        if(rnk[semi[low[v]]]>rnk[semi[x]]) low[v]=x;
        anc[v]=anc[anc[v]];
    }
    return low[v];
}
void dominators(int rt){
    for0(i,n){
        dom[i].clear(); rev[i].clear();
        rnk[i]=pre[i]=anc[i]=idom[i]=n;
        semi[i]=low[i]=i;
    }
    ord.clear(); dfspre(rt);
    for(int i=SZ(ord)-1;i;i--){
        int w=ord[i];
        for(int v:rev[w]){
            int u=eval(v);
            if(rnk[semi[w]]>rnk[semi[u]])semi[w]=semi[u];
        }
        dom[semi[w]].pb(w); anc[w]=pre[w];
        for(int v:dom[pre[w]]){
            int u=eval(v);
            idom[v]=(rnk[pre[w]]>rnk[semi[u]]?u:pre[w]);
        }
        dom[pre[w]].clear();
    }
```

```
    for(int w:ord) if(w!=rt&&idom[w]!=semi[w])
idom[w]=idom[idom[w]];
    fore(i,0,n) if(idom[i]==n)idom[i]=-1;
}
```

3.7 dsu on tree

```
vector<int> adj[N];
int L[N],V[N],sz[N],cur;

void dfs1(int u,int p){
    L[u] = cur, V[cur++] = u, sz[u] = 1;
    for (int v : adj[u]){
        if (v == p) continue;
        dfs1(v, u);
        sz[u] += sz[v];
    }
}

void dfs2(int u,int p,bool flag){
    int big = -1;
    for (int v : adj[u])
        if (v != p && (big == -1 || sz[big] < sz[v])) big =
v;
    for (int v : adj[u])
        if (v != p && v != big) dfs2(v, u, true);
    if (big != -1) dfs2(big, u, false);
    update(a[u], 1); // add
    for (int v : adj[u]){
        if (v == p || v == big) continue;
        for (int i=L[v]; i<L[v]+sz[v]; i++)
            update(a[V[i]], 1); // add
    }
    ans[u] = dif;
    if (!flag) return;
    for (int i=L[u]; i<L[u]+sz[u]; i++)
        update(a[V[i]], -1); // remove
} // by NekoRolly
```

3.8 floyd warshall

```
void Ini(int n) {
    for0(u, n) {
        for0(v, n) d[u][v] = INF, parent[u][v] = -1;
        d[u][u] = 0;
    }
}

void addEdge(int u, int v, int w) {
    if (w < d[u][v]) {
        d[u][v] = w;
        parent[u][v] = u;
    }
}

void floydWarshall(int n) { // O(V^3)
    for0(k, n) for0(u, n) for0(v, n) {
        if (d[u][k] == INF) continue;
        if (d[k][v] == INF) continue;
        if (d[u][k] + d[k][v] < d[u][v]) {
            d[u][v] = max(-INF, d[u][k] + d[k][v]);
            parent[u][v] = parent[k][v];
        }
    }

    // negative cycles
    for0(u, n) for0(v, n) for0(k, n) {
        if (d[k][k] < 0 && d[u][k] != INF && d[k][v] != INF)
{
            d[u][v] = -INF;
        }
    }
}
```

3.9 kuhn

```
// Maximum matching for bipartite unweighted graphs
struct Graph {
    int match[MX];
    bool vis[MX];
    vector<int> adj[MX];
    void addEdge(int u, int v);

    bool dfs(int u) {
        if (vis[u]) return false;
        vis[u] = true;
        for (int v : adj[u]) {
            if (match[v] == -1 || dfs(match[v])) {
                match[v] = u, match[u] = v;
                return true;
            }
        }
        return false;
    }

    int maximumMatching(int n) { // O(V * E)
        // If the bipartition is (V1, V2) we can iterate
from the side
        // that have less nodes (say is V1) and achieve O(V1
* E)
        fill(match, match + n, -1);
        int ans = 0;
        for (int u = 0; u < n; u++) {
            if (match[u] == -1) {
                fill(vis, vis + n, false);
                ans += dfs(u);
            }
        }
        return ans;
    }
} G;
```

3.10 lca

```
int tIn[MX], tOut[MX];
int anc[MX][LG];
int timer;

void dfs(int u) {
    tIn[u] = timer++;
    for (int j = 1; j < LG; j++) anc[u][j] = anc[anc[u][j -
1]][j - 1];
    for (int v : adj[u]) {
        if (v == anc[u][0]) continue;
        anc[v][0] = u;
        dfs(v);
    }
    tOut[u] = timer++;
}

bool isAnc(int u, int v) { // is u ancestor of v ?
    return tIn[u] <= tIn[v] && tOut[u] >= tOut[v];
}

int lca(int u, int v) { // O(log n)
    if (isAnc(u, v)) return u;
    if (isAnc(v, u)) return v;
    for (int i = LG - 1; i >= 0; i--) {
        if (!isAnc(anc[u][i], v)) u = anc[u][i];
    }
    return anc[u][0];
}
```

3.11 maxflow mincost

```
using Cap = long long;
using Cost = long long;

const int MX = 5000;
const Cap INF_CAP = 1e18;
const Cost INF_COST = 1e18;

struct Edge {
    int to;
    Cap flow, cap;
    Cost cost;
    int rev; // index of the backward edge in the adj list
of to
    Edge(int to, Cap cap, Cost cost, int rev)
        : to(to), flow(0), cap(cap), cost(cost), rev(rev) {}
};

struct Graph {
    vector<Edge> adj[MX];
```

```
    int parentEdge[MX];
    Cost pot[MX];
    //|pot[u]| <= the maximum sum of absolute cost in a path
    // which is also bounded by [(V - 1) * C]
    // where C is the maximum value of |cost(e)| for all
edges e
    void clear(int n) {
        for (int i = 0; i < n; i++) {
            adj[i].clear();
            pot[i] = 0;
        }
    }

    void addEdge(int u, int v, Cap w, Cost cost, bool dir) {
        adj[u].push_back(Edge(v, w, cost, adj[v].size()));
        adj[v].push_back(Edge(u, 0, -cost, adj[u].size() -
1));
        if (!dir) addEdge(v, u, w, cost, true);
    }

    void spfa(int s, int n) { // O(E V)
        vector<bool> inQueue(n, false);
        for (int i = 0; i < n; i++) pot[i] = INF_COST;
        queue<int> q;
        pot[s] = 0;
        inQueue[s] = true;
        q.push(s);
        while (!q.empty()) {
            int u = q.front();
            q.pop();
            inQueue[u] = false;
            for (Edge e : adj[u]) {
                int v = e.to;
                if (e.cap - e.flow > 0 && pot[u] + e.cost <
pot[v]) {
                    pot[v] = pot[u] + e.cost;
                    if (!inQueue[v]) q.push(v);
                    inQueue[v] = true;
                }
            }
        }
    }

    void pushFlow(int s, int t, Cap inc) {
        int v = t;
        while (v != s) {
            Edge &backward = adj[v][parentEdge[v]];
            int u = backward.to;
            Edge &forward = adj[u][backward.rev];
            forward.flow += inc;
            backward.flow -= inc;
            v = u;
        }
    }
```

```
    pair<Cap, Cost> dijkstra(int s, int t, int n) { // O(E
log V)
        //<flow, cost>
        using Path = pair<Cost, int>; //<weight, node>
        priority_queue<Path, vector<Path>, greater<Path>> q;
        vector<Cost> d(n, INF_COST);
        vector<Cap> residualCap(n, 0);
        d[s] = 0;
        residualCap[s] = INF_CAP;
        q.push(Path(d[s], s));
        while (!q.empty()) {
            auto [dist, u] = q.top();
            q.pop();
            if (dist != d[u]) continue;
            for (Edge e : adj[u]) {
                int v = e.to;
                Cap cf = e.cap - e.flow;
                Cost cost = e.cost + pot[u] - pot[v];
                if (cf > 0 && d[u] + cost < d[v]) {
                    d[v] = d[u] + cost;
                    q.push(Path(d[v], v));
                    residualCap[v] = min(residualCap[u],
cf);
                    parentEdge[v] = e.rev;
                }
            }
        }
        if (d[t] == INF_COST) return {0, 0};
        for (int i = 0; i < n; i++) {
            if (pot[i] != INF_COST) pot[i] += d[i];
        }
        Cap cf = residualCap[t];
        pushFlow(s, t, cf);
        return {cf, pot[t] * cf};
    }

    pair<Cap, Cost> minCostFlow(int s, int t, int n) {
        // O(E log V * maxFlow)
        // maxFlow <= V * U, where U is the maximum capacity
        // Assumption: Initially no negative cycles
        //<maxFlow, minCost>
        spfa(s, n); // not necessary if there is no negative
edges
        pair<Cap, Cost> inc;
        Cap flow = 0;
        Cost cost = 0;
        do {
            inc = dijkstra(s, t, n);
            flow += inc.first;
            cost += inc.second;
        } while (inc.first > 0);
        return {flow, cost};
    }
} G;
```

3.12 tree bridge

```
vector<int> adj[MX];
int tin[MX], low[MX], timer;
vector<pii> bridges;
int component[MX];
vector<int> nods[MX];
vector<array<int, 3>> adjcomp[MX];
void dfs(int u, int p) {
    tin[u] = low[u] = ++timer;
    for (int v : adj[u])
        if (v != p) {
            if (tin[v]) low[u] = min(low[u], tin[v]);
            else dfs(v, u), low[u] = min(low[u], low[v]);
            if (tin[u] < low[v]) bridges.pb({u, v});
        }
}
bool isbridge(int u, int v) {
    if (tin[u] > tin[v]) swap(u, v);
    return tin[u] < low[v];
}
void dfsc(int u, int comp) {
    component[u] = comp;
    nods[comp].pb(u);
    for (int v : adj[u])
        if (!isbridge(u, v) && !component[v]) dfsc(v, comp);
}
int tree_bridge(int n) {
    dfs(1, 1);
    int comp = 0;
    for (int u = 1; u <= n; u++)
        if (!component[u]) dfsc(u, ++comp);
    for (auto [u, v] : bridges) {
        int cu = component[u], cv = component[v];
        adjcomp[cu].pb({cv, u, v});
        adjcomp[cv].pb({cu, u, v});
    }
    return comp;
}
```

4 Math

4.1 berlekamp massey

```
const int MOD = 998244353;
struct mint {
    int v;
    mint(long long x = 0) : v(x % MOD) {
        if (v < 0) v += MOD;
    }
    mint operator+(mint o) const {
        return v + o.v >= MOD ? v + o.v - MOD : v + o.v;
    }
    mint operator-(mint o) const {
```

```
        return v - o.v < 0 ? v - o.v + MOD : v - o.v;
    }
    mint operator*(mint o) const {
        return 1LL * v * o.v % MOD;
    }
    mint pow(long long k) const {
        mint res = 1, base = *this;
        while (k) {
            if (k & 1) res = res * base;
            base = base * base;
            k >>= 1;
        }
        return res;
    }
    mint inv() const {
        return pow(MOD - 2);
    }
};

struct BM {
    vector<mint> base, c;
    // O(N^2) - Obtiene la recurrencia
    BM(vector<mint> s) : base(s) {
        int n = s.size(), L = 0, m = 0;
        vector<mint> C(n, 0), B(n, 0), T;
        C[0] = B[0] = 1;
        mint b = 1;
        for (int i = 0; i < n; i++) {
            m++;
            mint d = s[i];
            for (int j = 1; j <= L; j++) d = d + C[j] * s[i
- j];
            if (d.v == 0) continue;
            T = C;
            mint coef = d * b.inv();
            for (int j = m; j < n; j++) C[j] = C[j] - coef *
B[j - m];
            if (2 * L > i) continue;
            L = i + 1 - L, B = T, b = d, m = 0;
        }
        c.assign(C.begin() + 1, C.begin() + L + 1);
        for (mint &x : c) x = mint(0) - x;
    }

    // O(L^2 log K) - Obtiene el k-Ã©simo tÃ©rmino
    mint get(long long k) {
        int n = c.size();
        if (k < base.size()) return base[k];
        if (n == 0) return 0;
        auto combine = [&](vector<mint> A, vector<mint> B) {
            vector<mint> res(2 * n + 1, 0);
            for (int i = 0; i <= n; i++)
                for (int j = 0; j <= n; j++) res[i + j] =
res[i + j] + A[i] * B[j];
            for (int i = 2 * n; i > n; i--)
                for (int j = 0; j < n; j++) res[i - 1 - j] =
res[i - 1 - j] + res[i] * c[j];
            res.resize(n + 1);
            return res;
        };
        vector<mint> pol(n + 1, 0), e(n + 1, 0);
        pol[0] = 1;
        if (n > 0) e[1] = 1;
        k++;
        while (k) {
            if (k & 1) pol = combine(pol, e);
            e = combine(e, e);
            k >>= 1;
        }
        mint res = 0;
        for (int i = 0; i < n; i++) res = res + pol[i + 1] *
base[i];
        return res;
    }
};
```

4.2 cht

```
struct Line {
    mutable ll a, b, c;

    bool operator<(Line r) const {
        return a < r.a;
    }
    bool operator<(ll x) const {
        return c < x;
    }
};
// dynamically insert `a*x + b` lines and query for maximum
// at any x all operations have complexity O(log N)
struct LineContainer : multiset<Line, less<>> {
    ll div(ll a, ll b) {
        return a / b - ((a ^ b) < 0 && a % b);
    }

    bool isect(iterator x, iterator y) {
        if (y == end()) return x->c = INF, 0;
        if (x->a == y->a) x->c = x->b > y->b ? INF : -INF;
        else x->c = div(y->b - x->b, x->a - y->a);
        return x->c >= y->c;
    }

    void add(ll a, ll b) {
        // a *= -1, b *= -1 // for min
        auto z = insert({a, b, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y =
erase(y));
        while ((y = x) != begin() && (--x)->c >= y->c)
isect(x, erase(y));
    }

    ll query(ll x) {
        if (empty()) return -INF; // INF for min
        auto l = *lower_bound(x);
        return l.a * x + l.b;
        // return -l.a * x - l.b; // for min
    }
};
```

4.3 criba

```
bool criba[MX];
ll primerfactor[MX], phi[MX];
// criba para primerfactor, phi O(nlog(log(n)))
void completar() {
    for (int i = 0; i < MX; i++) {
        criba[i] = 1;
        primerfactor[i] = i;
        phi[i] = i;
    }
    criba[0] = 0;
    criba[1] = 0;
    for (ll i = 2; i < MX; i++) {
        if (criba[i]) {
            phi[i]--;
            for (ll j = 2 * i; j < MX; j += i) {
                criba[j] = 0;
                phi[j] -= phi[j] / i;
                if (primerfactor[j] == j) primerfactor[j] =
i;
            }
        }
    }
}
vector<pll> factorizacioncriba(ll n) { // O(log(n))
    vector<pll> ans;
    while (n > 1) {
        ll f = primerfactor[n];
        ll pot = 0;
        while (primerfactor[n] == f) {
            n /= f;
            pot++;
        }
        ans.pb({f, pot});
    }
    return ans;
}
vector<ll> divisorescriba(ll n) { // O(d(x))
    vector<ll> fact = {1};
    while (n > 1) {
        ll f = primerfactor[n];
        ll sz = fact.size();
        ll divisor = 1;
        while (n % f == 0) {
```

```
            divisor *= f;
            n /= f;
            rep(i, sz) fact.pb(fact[i] * divisor);
        }
    }
    return fact;
}
```

4.4 criba lineal

```
int spf[MX];
int mu[MX];
int phi[MX];
vector<int> primes;
void compute() {
    spf[1] = mu[1] = phi[1] = 1;
    for (int i = 2; i < MX; i++) {
        if (spf[i] == 0) {
            spf[i] = i;
            mu[i] = -1;
            phi[i] = i - 1;
            primes.pb(i);
        }
        for (int p : primes) {
            if (p > spf[i] || i * p >= MX) break;
            spf[i * p] = p;
            if (p == spf[i]) {
                mu[i * p] = 0;
                phi[i * p] = phi[i] * p;
            } else {
                mu[i * p] = -mu[i];
                phi[i * p] = phi[i] * (p - 1);
            }
        }
    }
}
```

4.5 diophantine

```
ll extended_gcd(ll a, ll b, ll &x, ll &y) {
    x = 1, y = 0;
    ll x1 = 0, y1 = 1;
    while (b) {
        ll q = a / b;
        tie(x, x1) = make_pair(x1, x - q * x1);
        tie(y, y1) = make_pair(y1, y - q * y1);
        tie(a, b) = make_pair(b, a - q * b);
    }
    return a < 0 ? (x *= -1, y *= -1, a *= -1, a) : a;
}
bool diophantine(ll a, ll b, ll c, ll &x, ll &y, ll &g) {
    if (a == 0 && b == 0) return c ? 0 : (x = y = 0, 1);
    g = extended_gcd(a, b, x, y);
    return (c % g == 0 ? (x *= c / g, y *= c / g, 1) : 0);
}
ll divdown(ll a, ll b) {
    return a / b - ((a ^ b) < 0 && a % b);
}
ll divup(ll a, ll b) {
    return a / b + ((a ^ b) >= 0 && a % b);
}
vector<pll> allsolution(ll a, ll b, ll c, ll lx, ll rx, ll
ly, ll ry) {
    // x = x0 + k*(b/g)    y = y0 - k*(a/g)
    ll x0, y0, g;
    if (!diophantine(a, b, c, x0, y0, g)) return {};
    vector<pll> ans;
    if (!g) {
        for (ll x = lx; x <= rx; x++)
            for (ll y = ly; y <= ry; y++) ans.pb({x, y});
        return ans;
    }
    if (!a) {
        ll y = c / b;
        if (!(ly <= y && y <= ry)) return {};
        for (ll x = lx; x <= rx; x++) ans.pb({x, y});
        return ans;
    }
    if (!b) {
        ll x = c / a;
        if (!(lx <= x && x <= rx)) return {};
        for (ll y = ly; y <= ry; y++) ans.pb({x, y});
        return ans;
    }
    ll dx = b / g, dy = a / g;
    ll kmin, kmax;
    if (dx > 0) kmin = divup(lx - x0, dx), kmax = divdown(rx
- x0, dx);
    else kmin = divup(rx - x0, dx), kmax = divdown(lx - x0,
dx);
    if (dy > 0) kmin = max(kmin, divup(y0 - ry, dy)), kmax =
min(kmax, divdown(y0 - ly, dy));
    else kmin = max(kmin, divup(y0 - ly, dy)), kmax =
min(kmax, divdown(y0 - ry, dy));
    for (ll k = kmin; k <= kmax; k++) ans.pb({x0 + k * dx,
y0 - k * dy});
    return ans;
}
bool positive_sol_minf(ll a, ll b, ll c, ll &x, ll &y, ll
cx, ll cy) {
    // f(x,y)= cx*x + cy*y
    ll g;
    if (!diophantine(a, b, c, x, y, g)) return 0;
    ll dx = b / g, dy = a / g;
    ll kmin = -INF, kmax = INF;
    dx > 0 ? kmin = max(kmin, divup(-x, dx)) : kmax =
min(kmax, divdown(-x, dx));
    dy < 0 ? kmin = max(kmin, divup(y, dy)) : kmax =
min(kmax, divdown(y, dy));
    if (kmin > kmax) return 0;
    ll df = cx * dx - cy * dy;
    ll k = df >= 0 ? kmin : kmax; // df<0 para max
    x += dx * k;
    y -= dy * k;
    return 1;
}
```

4.6 dirichlet convolution

```
ll pref[B];
unordered_map<ll, __int128> dp;
vector<int> primes;
void compute() {
    pref[1] = 1;
    for (int i = 2; i < B; i++) {
        if (pref[i] == 0) {
            pref[i] = i - 1;
            primes.pb(i);
        }
        for (int p : primes) {
            if (i * p >= B) break;
            if (i % p == 0) {
                pref[i * p] = pref[i] * p;
                break;
            } else pref[i * p] = pref[i] * (p - 1);
        }
    }
    for (int i = 1; i < B; i++) pref[i] += pref[i - 1];
}
// Formula:S_f(n) = (1 / g(1)) * (S_f*g(n) - sum( S_f(q) *
( S_g(r) - S_g(l-1) ) )
// q = n/l
// r = n/q
// sum : for(ll l = 2, r; l<=n; l = r+1)
// para f=phi, g=1: phi * 1 = ID
__int128 S_f(ll n) {
    if (n < B) return pref[n];
    if (dp.count(n)) return dp[n];
    __int128 ans = (__int128)n * (n + 1) / 2;
    for (ll l = 2, r; l <= n; l = r + 1) {
        ll q = n / l;
        r = n / q;
        ans -= S_f(q) * (r - l + 1);
    }
    return dp[n] = ans;
}
```

4.7 fft

```
typedef complex<double> cd;
vector<vector<cd>> w;
void calc(int log) {
    w.resize(log + 1);
    w[0].resize(1, 1);
    for (int l = 1, len = 2; l <= log; l++, len <<= 1) {
```

```
        w[l].resize(len / 2);
        cd wn(cos(2 * PI / len), sin(2 * PI / len));
        for (int j = 0; 2 * j < len; j++)
            w[l][j] = (j & 1 ? w[l - 1][j / 2] * wn : w[l -
1][j / 2]);
    }
}
void fft(vector<cd> &a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int l = 1, len = 2; len <= n; l++, len <<= 1)
        for (int i = 0; i < n; i += len)
            for (int j = 0; 2 * j < len; j++) {
                cd u = a[i + j], v = a[i + j + len / 2] *
w[l][j];
                a[i + j] = u + v;
                a[i + j + len / 2] = u - v;
            }
    if (invert) {
        reverse(a.begin() + 1, a.end());
        for (int i = 0; i < n; i++) a[i] /= n;
    }
}
vector<ll> polymult(vector<ll> &a, vector<ll> &b) {
    int n = 1, log = 0;
    while (n < a.size() + b.size() - 1) n <<= 1, log++;
    calc(log);
    vector<cd> A(n);
    for (int i = 0; i < n; i++) A[i] = cd(i < a.size() ?
a[i] : 0, i < b.size() ? b[i] : 0);
    fft(A, 0);
    for (int i = 0; i < n; i++) A[i] *= A[i];
    fft(A, 1);
    vector<ll> ans(a.size() + b.size() - 1);
    for (int i = 0; i < ans.size(); i++) ans[i] =
round(A[i].imag() / 2);
    return ans;
}
// 2D-FFT
void fft2D(vector<vector<cd>> &a, bool invert) {
    int n = a.size(), m = a[0].size();
    for (int i = 0; i < n; i++) fft(a[i], invert);
    vector<cd> b(n);
    for (int j = 0; j < m; j++) {
        for (int i = 0; i < n; i++) b[i] = a[i][j];
        fft(b, invert);
        for (int i = 0; i < n; i++) a[i][j] = b[i];
    }
}
vector<vector<ll>> convolution2D(vector<vector<ll>> &a,
vector<vector<ll>> &b) {
    int n = 1, logn = 0;
    while (n < a.size() + b.size() - 1) n <<= 1, logn++;
    int m = 1, logm = 0;
    while (m < a[0].size() + b[0].size() - 1) m <<= 1, logm+
+;
    calc(max(logn, logm));
    vector<vector<cd>> A(n, vector<cd>(m));
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            A[i][j] = cd(i < a.size() && j < a[0].size() ?
a[i][j] : 0,
                         i < b.size() && j < b[0].size() ?
b[i][j] : 0);
    fft2D(A, 0);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++) A[i][j] *= A[i][j];
    fft2D(A, 1);
    vector<vector<ll>> ans(a.size() + b.size() - 1,
vector<ll>(a[0].size() + b[0].size() - 1));
    for (int i = 0; i < ans.size(); i++)
        for (int j = 0; j < ans[0].size(); j++) ans[i][j] =
round(A[i][j].imag() / 2);
    return ans;
}
```

4.8 fwht bitwise convolution

```
//OR-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <<= 1, h <<= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) a[j + h] +=
a[j];
}
void ifwht(vector<ll> &A) {
    int n = A.size();
    for (int s = n, h = n >> 1; s >= 2; s >>= 1, h >>= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) A[j + h] -=
A[j];
}
vector<ll> OrConvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <<= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}
vector<ll> SubsetConvolution(vector<ll> a, vector<ll> b) {
    int n = 1, bit = 1;
    while (n < a.size() || n < b.size()) n <<= 1, bit++;
    a.resize(n), b.resize(n);
    vector<vector<ll>> A(bit + 1, vector<ll>(n)), B(bit + 1,
vector<ll>(n));
    for (int i = 0; i < n; i++) {
        int f = __builtin_popcount(i);
        A[f][i] = a[i], B[f][i] = b[i];
    }
    for (int i = 0; i <= bit; i++) fwht(A[i]), fwht(B[i]);
    vector<ll> conv(n, 0);
    for (int k = 0; k < n; k++) {
        int f = __builtin_popcount(k);
        for (int i = 0; i <= f; i++) conv[k] += A[i][k] *
B[f - i][k];
    }
    ifwht(conv);
    return conv;
}
//AND-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <<= 1, h <<= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) a[j] += a[j +
h];
}
void ifwht(vector<ll> &A) {
    int n = A.size();
    for (int s = n, h = n >> 1; s >= 2; s >>= 1, h >>= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) A[j] -= A[j +
h];
}
vector<ll> Andconvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <<= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}
//XOR-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <<= 1, h <<= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) {
                ll a1 = a[j + h], a0 = a[j];
                a[j + h] = a0 - a1, a[j] = a0 + a1;
            }
}
void ifwht(vector<ll> &A) {
    fwht(A);
```

```
    int n = A.size();
    for (int i = 0; i < n; i++) A[i] /= n;
}
vector<ll> Xorconvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <<= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}
```

4.9 gauss

```
const int N = 3e5 + 9;
const double eps = 1e-9;
int Gauss(vector<vector<double>> a, vector<double> &ans) {
    int n = (int)a.size(), m = (int)a[0].size() - 1;
    vector<int> pos(m, -1);
    double det = 1;
    int rank = 0;
    for (int col = 0, row = 0; col < m && row < n; ++col) {
        int mx = row;
        for (int i = row; i < n; i++)
            if (fabs(a[i][col]) > fabs(a[mx][col])) mx = i;
        if (fabs(a[mx][col]) < eps) {
            det = 0;
            continue;
        }
        for (int i = col; i <= m; i++) swap(a[row][i], a[mx]
[i]);
        if (row != mx) det = -det;
        det *= a[row][col];
        pos[col] = row;
        for (int i = 0; i < n; i++) {
            if (i != row && fabs(a[i][col]) > eps) {
                double c = a[i][col] / a[row][col];
                for (int j = col; j <= m; j++) a[i][j] -=
a[row][j] * c;
            }
        }
        ++row;
        ++rank;
    }
    ans.assign(m, 0);
    for (int i = 0; i < m; i++) {
        if (pos[i] != -1) ans[i] = a[pos[i]][m] / a[pos[i]]
[i];
    }
    for (int i = 0; i < n; i++) {
        double sum = 0;
        for (int j = 0; j < m; j++) sum += ans[j] * a[i][j];
        if (fabs(sum - a[i][m]) > eps) return -1; // no
solution
    }
    for (int i = 0; i < m; i++)
        if (pos[i] == -1) return 2; // infinte solutions
    return 1;                       // unique solution
}
```

4.10 geo2d

```
struct point {
    ll x, y;
    point() {}
    point(ll _x, ll _y) : x(_x), y(_y) {}
    point operator+(const point &p) const {
        return {x + p.x, y + p.y};
    }
    point operator-(const point &p) const {
        return {x - p.x, y - p.y};
    }
    bool operator==(const point &p) const {
        return x == p.x && y == p.y;
    }
    bool operator<(const point &p) const {
        if (x == p.x) return y < p.y;
        return x < p.x;
    }
};
// Producto punto
long long dot(const point &a, const point &b) {
    return a.x * b.x + a.y * b.y;
}
// Producto cruzado (determinante)
long long cross(const point &a, const point &b) {
    return a.x * b.y - a.y * b.x;
}
// Cross entre vectores AB y AC
long long cross(const point &a, const point &b, const point
&c) {
    return (b.x - a.x) * (c.y - a.y) - (b.y - a.y) * (c.x -
a.x);
}
// El punto p estÃ¡ en el segmento ab?
bool onSegment(const point &a, const point &b, const point
&p) {
    if (cross(a, b, p) != 0) return false;
    return min(a.x, b.x) <= p.x && p.x <= max(a.x, b.x) &&
min(a.y, b.y) <= p.y &&
           p.y <= max(a.y, b.y);
}
// Interseccion de 2 segmentos
bool intersect(const point &p1, const point &q1, const point
&p2, const point &q2) {
    long long d1 = cross(p1, q1, p2);
    long long d2 = cross(p1, q1, q2);
    long long d3 = cross(p2, q2, p1);
    long long d4 = cross(p2, q2, q1);
    if ((d1 > 0 && d2 < 0 || d1 < 0 && d2 > 0) && (d3 > 0 &&
d4 < 0 || d3 < 0 && d4 > 0))
        return true;
    if (d1 == 0 && onSegment(p1, q1, p2)) return true;
    if (d2 == 0 && onSegment(p1, q1, q2)) return true;
    if (d3 == 0 && onSegment(p2, q2, p1)) return true;
    if (d4 == 0 && onSegment(p2, q2, q1)) return true;
    return false;
}
bool pointInPolygon(const vector<point> &poly, const point
&p) {
    bool inside = false;
    int n = poly.size();
    for (int i = 0, j = n - 1; i < n; j = i++) {
        const point &a = poly[i];
        const point &b = poly[j];
        if (onSegment(a, b, p)) return true;
        if ((a.y > p.y) != (b.y > p.y)) {
            long long x_intersection = (b.x - a.x) * (p.y -
a.y) - (b.y - a.y) * (p.x - a.x);
            long long y_diff = b.y - a.y;
            bool right = (x_intersection * y_diff) < 0;
            if (right) inside = !inside;
        }
    }
    return inside;
}
```

4.11 matrix

```
#define matrix array<array<int, N>, N>
matrix multiply(matrix &a, matrix &b) {
    matrix ans{};
    for (int i = 0; i < N; i++)
        for (int j = 0; j < N; j++)
            for (int k = 0; k < N; k++) ans[i][j] =
smod(ans[i][j], mult(a[i][k], b[k][j]));
    return ans;
}
matrix pow(matrix &a, ll e) {
    matrix ans{};
    for (int i = 0; i < N; i++) ans[i][i] = 1;
    while (e) {
        if (e & 1) ans = multiply(ans, a);
        a = multiply(a, a);
        e >>= 1;
    }
    return ans;
}
```

4.12 millerrabin polardrho

```
ll mult(ll a, ll b, ll mod) {
    return (int128)a * b % mod;
```

```
}
ll qp(ll a, ll e, ll mod) {
    ll ans = 1;
    while (e) {
        if (e & 1) ans = mult(ans, a, mod);
        a = mult(a, a, mod);
        e >>= 1;
    }
    return ans;
}
bool compositetest(ll a, ll d, ll s, ll n) {
    ll x = qp(a, d, n);
    if (x == 1 || x == n - 1) return 0;
    for (ll i = 0; i < s - 1; i++) {
        x = mult(x, x, n);
        if (x == n - 1) return 0;
    }
    return 1;
}
bool esprimo(ll n) {
    if (n < 2) return 0;
    ll d = n - 1, s = 0;
    while ((d & 1) == 0) d >>= 1, s++;
    for (ll a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,
37}) {
        if (n == a) return 1;
        if (compositetest(a, d, s, n)) return 0;
    }
    return 1;
}
mt19937_64
rng(chrono::steady_clock::now().time_since_epoch().count());
ll random(ll a, ll b) {
    return uniform_int_distribution<ll>(a, b)(rng);
}
ll f(ll x, ll c, ll mod) {
    ll m = mult(x, x, mod);
    return m + c >= mod ? m + c - mod : m + c;
}
ll getfactor(ll n) {
    if (n % 2 == 0) return 2;
    if (esprimo(n)) return n;
    while (true) {
        ll y = random(1, n - 1);
        ll c = random(1, n - 1);
        ll m = 64, g = 1, r = 1, q = 1;
        ll x, ys;
        while (g == 1) {
            x = y;
            for (ll i = 0; i < r; i++) y = f(y, c, n);
            ll k = 0;
            while (k < r && g == 1) {
                ys = y;
                ll lim = min(m, r - k);
                for (ll i = 0; i < lim; i++) {
                    y = f(y, c, n);
                    ll diff = x > y ? x - y : y - x;
                    q = mult(q, diff, n);
                }
                g = __gcd(q, n);
                k += m;
            }
            r <<= 1;
        }
        if (g == n) {
            do {
                ys = f(ys, c, n);
                ll diff = x > ys ? x - ys : ys - x;
                g = __gcd(diff, n);
            } while (g == 1);
        }

        if (g > 1 && g < n) return g;
    }
}
vector<pll> fp(ll n) {
    stack<ll> st;
    st.push(n);
    vector<pll> ans;
    map<ll, ll> cont;
    while (!st.empty()) {
        ll x = st.top();
        st.pop();
        if (x == 1) continue;
        if (esprimo(x)) cont[x]++;
        else {
            ll f = getfactor(x);
            st.push(f), st.push(x / f);
        }
    }
    for (auto [x, f] : cont) ans.pb({x, f});
    return ans;
}
```

4.13 ntt

```
const ll MOD = 998244353;
vector<vector<int>> w;
int pow2, root;
int smod(int a, int b, int mod) {return (a + b >= mod ? a +
b - mod : a + b);}
int rmod(int a, int b, int mod) {return (a - b < 0 ? a - b +
mod : a - b);}
int mult(int a, int b, int mod) {return (ll)a * b % mod;}
int qp(int a, int e, int mod) {
    int ans = 1;
    while (e) {
        if (e & 1) ans = mult(ans, a, mod);
        a = mult(a, a, mod);
        e >>= 1;
    }
    return ans;
}
void calc(int log, int mod) {
    w.resize(log + 1);
    w[0].resize(1, 1);
    for (int l = 1, len = 2; l <= log; l++, len <<= 1) {
        w[l].resize(len / 2);
        int wn = qp(root, pow2 / len, mod);
        for (int j = 0; 2 * j < len; j++) {
            if (j & 1) w[l][j] = mult(w[l - 1][j / 2], wn,
mod);
            else w[l][j] = w[l - 1][j / 2];
        }
    }
}
void findroot(int mod) {
    pow2 = 1;
    int k = mod - 1;
    while (!(k & 1)) pow2 <<= 1, k >>= 1;
    root = 1;
    while (qp(root, pow2, mod) != 1 || qp(root, pow2 / 2,
mod) == 1) root++;
}
void ntt(vector<int> &a, bool invert, int mod) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int l = 1, len = 2; len <= n; l++, len <<= 1)
        for (int i = 0; i < n; i += len)
            for (int j = 0; 2 * j < len; j++) {
                int u = a[i + j], v = mult(a[i + j + len /
2], w[l][j], mod);
                a[i + j] = smod(u, v, mod);
                a[i + j + len / 2] = rmod(u, v, mod);
            }
    if (invert) {
        reverse(a.begin() + 1, a.end());
        int invn = qp(n, mod - 2, mod);
        for (int i = 0; i < n; i++) a[i] = mult(a[i], invn,
mod);
    }
}
vector<int> convolution(vector<int> a, vector<int> b, int
mod = MOD) {
    int n = 1, log = 0, rn = a.size() + b.size() - 1;
    while (n < a.size() + b.size() - 1) n <<= 1, log++;
    findroot(mod);
    calc(log, mod);
    a.resize(n), b.resize(n);
    ntt(a, 0, mod), ntt(b, 0, mod);
```

```
    for (int i = 0; i < n; i++) a[i] = mult(a[i], b[i],
mod);
    ntt(a, 1, mod);
    a.resize(rn);
    return a;
}
// 2D-FFT
void ntt2D(vector<vector<int>> &a, bool invert, int mod) {
    int n = a.size(), m = a[0].size();
    for (int i = 0; i < n; i++) ntt(a[i], invert, mod);
    vector<int> b(n);
    for (int j = 0; j < m; j++) {
        for (int i = 0; i < n; i++) b[i] = a[i][j];
        ntt(b, invert, mod);
        for (int i = 0; i < n; i++) a[i][j] = b[i];
    }
}
vector<vector<int>> convolution2D(vector<vector<int>> a,
vector<vector<int>> b,
                                  int mod = MOD) {
    int n = 1, logn = 0, rn = a.size() + b.size() - 1;
    while (n < rn) n <<= 1, logn++;
    int m = 1, logm = 0, rm = a[0].size() + b[0].size() - 1;
    while (m < rm) m <<= 1, logm++;
    findroot(mod);
    calc(max(logn, logm), mod);
    a.resize(n), b.resize(n);
    for (int i = 0; i < n; i++) a[i].resize(m),
b[i].resize(m);
    ntt2D(a, 0, mod), ntt2D(b, 0, mod);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++) a[i][j] = mult(a[i][j],
b[i][j], mod);
    ntt2D(a, 1, mod);
    a.resize(rn);
    for (int i = 0; i < rn; i++) a[i].resize(rm);
    return a;
}
```

4.14 online fft

```
// A(DP) X G_STATIC
// EcuaciÃ³n: dp[i] += c[i] * sum_{j} (A[j] * G[i - j -
off])
void dfs_AG(int l, int r, int offset, vector<int> &dp,
vector<int> &A, const vector<int> &G,
            const vector<int> &c) {
    if (l == r) {
        // Aqui ya esta calculado dp[i] = sum_{j} (A[j] *
G[i - j - off]) y dp[j]
        // para j < i ya esta calculado
        dp[l] = mult(dp[l], c[l]);
        A[l] = mult(l, dp[l]);
        return;
    }
    int mid = (l + r) >> 1;
    dfs_AG(l, mid, offset, dp, A, G, c);
    vector<int> a(A.begin() + l, A.begin() + mid + 1);
    vector<int> b(G.begin(), G.begin() + max(0, r - l -
offset + 1));
    vector<int> conv = convolution(a, b);
    for (int i = max(mid + 1, l + offset); i <= min(r,
(int)conv.size() + l + offset - 1); i++)
        dp[i] = smod(dp[i], conv[i - l - offset]);
    dfs_AG(mid + 1, r, offset, dp, A, G, c);
}

// A(DP) X A(DP)
// EcuaciÃ³n SimÃ©trica: dp[i] = c[i] * sum_{j} (A[j] * A[i
- j - off]) + g[i]
// Ejemplo de dependencia: A[i] = i * dp[i]
void dfs_sim(int l, int r, int offset, vector<int> &dp,
vector<int> &A, const vector<int> &c,
             const vector<int> &g) {
    if (l == r) {
        dp[l] = smod(mult(c[l], dp[l]), g[l]);
        A[l] = mult(l, dp[l]);
        return;
    }
    int mid = (l + r) >> 1;
    dfs_sim(l, mid, offset, dp, A, c, g);
    vector<int> a(A.begin() + l, A.begin() + mid + 1);
    vector<int> b =
        (l == 0) ? a : vector<int>(A.begin(), A.begin() +
max(0, r - l - offset + 1));
    vector<int> conv = convolution(a, b);
    for (int i = max(mid + 1, l + offset); i <= min(r,
(int)conv.size() + l + offset - 1); i++)
        dp[i] = smod(dp[i], mult(conv[i - l - offset], 2 - !
l));
    dfs_sim(mid + 1, r, offset, dp, A, c, g);
}

// A(DP) X B(DP)
// EcuaciÃ³n: dp[i] += c[i] * sum_{j} (A[j] * B[i - j -
off])
// A[i] = i * dp[i]
// B[i] = sum_{k=0}^{i} dp[k] (Suma prefija)
void dfs_gen(int l, int r, int offset, vector<int> &dp,
vector<int> &A, vector<int> &B,
             const vector<int> &c) {
    if (l == r) {
        dp[l] = mult(dp[l], c[l]); // dp[l] = c[i] * sum_{j}
(A[j] * B[i - j - off])
        A[l] = mult(l, dp[l]);
        B[l] = smod((l > 0) ? B[l - 1] : 0, dp[l]);
        return;
    }
    int mid = (l + r) >> 1;
    dfs_gen(l, mid, offset, dp, A, B, c);
    vector<int> a1(A.begin() + l, A.begin() + mid + 1);
    vector<int> b1(B.begin(), B.begin() + max(0, r - l -
offset + 1));
    vector<int> conv1 = convolution(a1, b1);
    for (int i = max(mid + 1, l + offset); i <= min(r,
(int)conv1.size() + l + offset - 1);
         i++)
        dp[i] = smod(dp[i], conv1[i - l - offset]);
    if (l > 0) {
        vector<int> a2(B.begin() + l, B.begin() + mid + 1);
        vector<int> b2(A.begin(), A.begin() + max(0, r - l -
offset + 1));
        vector<int> conv2 = convolution(a2, b2);
        for (int i = max(mid + 1, l + offset); i <= min(r,
(int)conv2.size() + l + offset - 1);
             i++)
            dp[i] = smod(dp[i], conv2[i - l - offset]);
    }
    dfs_gen(mid + 1, r, offset, dp, A, B, c);
}
```

4.15 primitive root

```
int primitive_root(int p) {
    int n = p - 1; // n=phi(p) -> n=p-1 si p es primo
    vector<int> fp;
    for (int i = 2; i * i <= n; i++)
        if (n % i == 0) {
            fp.pb(i);
            while (n % i == 0) n /= i;
        }
    if (n > 1) fp.pb(n);
    n = p - 1;
    bool ok = 0;
    for (int g = 2; g <= n; g++) {
        bool ok = 1;
        for (int x : fp)
            if (qp(g, n / x, p) == 1) {
                ok = 0;
                break;
            }
        if (ok) return g;
    }
    return 0;
}
```

4.16 random

```
static mt19937 rng(chrono::steady_clock::now()
    .time_since_epoch().count());
#define rnd(a,b) (uniform_int_distribution<ll>(a,b)(rng))
```

4.17 theorems

4.17.1 Combinatoria

4.17.1.1 General

- $\sum_{k=0}^{n}\binom{n-k}{k} = \text{Fib}_{n+1}$
- $\binom{n}{k} = \binom{n}{n-k}$
- $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$
- $k\binom{n}{k} = n\binom{n-1}{k-1}$
- $\sum_{i=0}^{n}\binom{n}{i} = 2^n$

Identidad de Vandermonde:

$$\sum_{k=0}^{r}\binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$$

Identidad del Palo de Hockey (Hockey-Stick):

$$\sum_{i=r}^{n}\binom{i}{r} = \binom{n+1}{r+1}$$

4.17.1.2 TriÃ¡ngulo de Pascal

- En una fila p donde p es un nÃºmero primo, todos los tÃ©rminos en esa fila excepto los unos son mÃºltiplos de p.
- **Teorema de Kummer**: Para enteros $n \geq m \geq 0$ y un nÃºmero primo p, la mayor potencia de p que divide a $\binom{n}{m}$ es igual al nÃºmero de acarreos cuando m se suma a $n-m$ en base p.

4.17.1.3 NÃºmeros de Catalan

- FÃ³rmula general: $C_n = \frac{1}{n+1}\binom{2n}{n}$
- Recurrencia: $C_0 = 1, C_1 = 1$ y $C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$

Representa, entre otras cosas, el nÃºmero de secuencias de parÃ©ntesis balanceadas o triangulaciones de un polÃgono convexo.

4.17.1.4 NÃºmeros de Stirling (Primera y Segunda Especie)

- **Primera especie** (nÃºmero de permutaciones con k ciclos disjuntos):
  $S(n,k) = (n-1)\cdot S(n-1,k) + S(n-1,k-1)$
- **Segunda especie** (maneras de particionar un conjunto de n objetos en k subconjuntos no vacÃos):
  $S(n,k) = k\cdot S(n-1,k) + S(n-1,k-1)$

4.17.1.5 NÃºmeros de Bell

Cuenta el nÃºmero total de particiones posibles de un conjunto.

$$B_{n+1} = \sum_{k=0}^{n}\binom{n}{k}\cdot B_k$$

4.17.2 MatemÃ¡ticas

4.17.2.1 FÃ³rmulas Generales

- $a^k - b^k = (a-b)\cdot\left(a^{k-1}b^0 + a^{k-2}b^1 + ... + a^0 b^{k-1}\right)$
- $|a-b| + |b-c| + |c-a| = 2(\max(a,b,c) - \min(a,b,c))$

Identidad de Lagrange:

$$\left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} b_k^2\right) - \left(\sum_{k=1}^{n} a_k b_k\right)^2 = \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\left(a_i b_j - a_j b_i\right)^2$$

4.17.2.2 FÃ³rmulas de Vieta

Para un polinomio de grado n: $p(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_0$ con raÃces $r_1, r_2, ..., r_n$:

- $r_1 + r_2 + ... + r_n = -\frac{a_{n-1}}{a_n}$
- $r_1 r_2 ... r_n = (-1)^n \frac{a_0}{a_n}$

4.17.2.3 SucesiÃ³n de Fibonacci

- $F_n = \sum_{k=0}^{\lfloor\frac{n-1}{2}\rfloor}\binom{n-k-1}{k}$
- **FÃ³rmula de Binet:** $F_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$
- $\sum_{i=1}^{n} F_i = F_{n+2} - 1$
- $\gcd(F_m, F_n) = F_{\gcd(m,n)}$

4.17.2.4 Ternas PitagÃ³ricas

FÃ³rmula de Euclides para generar ternas donde $a^2 + b^2 = c^2$: Para $m > n > 0$:

$$a = m^2 - n^2, \quad b = 2mn, \quad c = m^2 + n^2$$

4.17.3 TeorÃa de NÃºmeros

4.17.3.1 FunciÃ³n Totiente de Euler (φ)

- $\varphi(n) = n\prod_{p\mid n}\left(1 - \frac{1}{p}\right)$ donde p son factores primos distintos.
- Si a y b son coprimos, entonces $\varphi(a\cdot b) = \varphi(a)\cdot\varphi(b)$
- $\sum_{d\mid n}\varphi(d) = n$

4.17.3.2 FunciÃ³n e InversiÃ³n de MÃ¶bius

$\mu(n)$ toma valores en $\{-1, 0, 1\}$:

- 1 si es libre de cuadrados con cantidad par de factores primos.
- -1 si es libre de cuadrados con cantidad impar de factores primos.
- 0 si contiene algÃºn factor primo al cuadrado.

Teorema de inversiÃ³n de MÃ¶bius: Si $g(n) = \sum_{d\mid n} f(d)$, entonces:

$$f(n) = \sum_{d\mid n}\mu(d)g\left(\frac{n}{d}\right)$$

4.17.3.3 MCD (GCD) y MCM (LCM)

- $\gcd(a,b)\cdot\text{lcm}(a,b) = |a\cdot b|$
- $\gcd(a,\text{lcm}(b,c)) = \text{lcm}(\gcd(a,b),\gcd(a,c))$
- $\sum_{k=1}^{n}\gcd(k,n) = \sum_{d\mid n} d\cdot\varphi\left(\frac{n}{d}\right)$

4.17.4 Operaciones de Bits (Bitwise) y MiscelÃ¡nea

- $a + b = (a \oplus b) + 2(a \text{ AND } b)$
- $a + b = (a \mid b) + (a \text{ AND } b)$
- El k-Ã©simo bit estÃ¡ encendido en x si y solo si $x \bmod 2^k \geq 2^{k-1}$.
- $1 \oplus 2 \oplus 3 \oplus ... \oplus (4k-1) = 0$ para todo $k \geq 0$.

4.17.4.1 Teorema de ErdÅ‘sâ€“Gallai

Una secuencia descendente de grados $d_1 \geq d_2 \geq ... \geq d_n$ puede representar un grafo simple finito si la suma total de grados es par y:

$$\sum_{i=1}^{k} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(d_i, k)$$

para cada $1 \leq k \leq n$

4.18 xor basis

```
template <typename T, int LOG>
struct Basis {
    T basis[LOG] = {0};
    int dim = 0;
    bool insert(T x) { // FormarÃ¡ parte del Basis?
        for (int i = LOG - 1; i >= 0; i--)
            if ((x >> i) & 1) {
                if (basis[i]) x ^= basis[i];
                else {
                    basis[i] = x, dim++;
                    return true;
                }
            }
        return false;
    }
    T size() {
        return ((T)1 << dim);
    }
    bool find(T x) { // Â¿x â^^ span{x1,x2...}?
        for (int i = LOG - 1; i >= 0; i--) x = min(x, x ^
basis[i]);
        return x == 0;
    }
    T max_xor(T x = 0) { // max (x^a / a â^^ span{x1,x2...})
        for (int i = LOG - 1; i >= 0; i--) x = max(x, x ^
basis[i]);
        return x;
    }
    T kth(T k) { // The k_smaller of span{x1,x2...}
        if (k < 1 || k > ((T)1 << dim)) return -1;
        T x = 0, cont = ((T)1 << dim);
        for (int i = LOG - 1; i >= 0; i--)
            if (basis[i]) {
                if (k > (cont >> 1)) {
                    if (!((x >> i) & 1)) x ^= basis[i];
                    k -= (cont >> 1);
                } else if ((x >> i) & 1) x ^= basis[i];
                cont >>= 1;
            }
        return x;
    }
}
```

5 Strings

5.1 aho corasick

```
struct Aho_Corasick{
    int t[N][K],cur;
    int pa[N],go[N][K],lnk[N];
    queue<int> qu;

    int insert(string &s){
        int u = 0;
        for (char &c : s){
            int k = c-'a';
            if (t[u][k] == 0)
                t[u][k] = ++cur, pa[cur] = u;
            u = t[u][k];
        }
        return u;
    }

    void bfs(){
        fill(lnk+1, lnk+cur+1, -1);
        for (qu.push(0); !qu.empty(); ){
            int u = qu.front(); qu.pop();
            for (int k=0; k<K; k++){
                int v = t[u][k];
                if (v == 0) go[u][k] = go[lnk[u]][k];
                else{
                    lnk[v] = u == 0 ? 0 : go[lnk[u]][k];
                    go[u][k] = t[u][k];
                    qu.push(v);
                }
            }
        }
    }
} aho; // by NekoRolly
```

5.2 hashing

```
struct Hashing{
    const int mod = 1e9+9;
    const int p = 367, q = 467;
    int ep[N],eq[N],prp[N],prq[N];

    int P(int a,int b){ return 1ll*a*b%mod;}
    int S(int a,int b){ return (a+b)%mod;}

    void build(string &s){
        int n = s.size();
        ep[0] = eq[0] = 1;
        prp[0] = prq[0] = 0;
        for (int i=1; i<=n; i++){
            prp[i] = S(P(prp[i-1], p), s[i-1]);
            prq[i] = S(P(prq[i-1], q), s[i-1]);
            ep[i] = P(ep[i-1], p);
            eq[i] = P(eq[i-1], q);
        }
    }

    ll substr(int i,int len){ i++;
        ll hp = S(prp[i+len-1], mod-P(prp[i-1], ep[len]));
        ll hq = S(prq[i+len-1], mod-P(prq[i-1], eq[len]));
        return hp*mod + hq;
    }
} H1, H2; // by NekoRolly
```

5.3 manacher

```
// d1[i] odd number of palindromes centered at i
// d2[i] even number of palindromes centered at (i, i + 1)
void manacher(string &s, vector<int> &d1, vector<int> &d2) {
    int n = s.size();
    d1 = vector<int>(n);
    d2 = vector<int>(n, 0);
    int l = -1, r = -1;
    for (int i = 0; i < n; i++) {
        d1[i] = (i > r) ? 1 : min(d1[l + r - i], r - i + 1);
        while (i - d1[i] >= 0 && i + d1[i] < n && s[i -
d1[i]] == s[i + d1[i]]) {
            d1[i]++;
        }
        if (i + d1[i] - 1 > r) r = i + d1[i] - 1, l = i -
(d1[i] - 1);
    }
    l = -1, r = -1;
    for (int i = 0; i < n - 1; i++) {
        d2[i] = (i >= r) ? 0 : min(d2[l + r - i - 1], r -
i);
        while (i - d2[i] >= 0 && i + d2[i] + 1 < n && s[i -
d2[i]] == s[i + d2[i] + 1]) {
            d2[i]++;
        }
        if (i + d2[i] > r && d2[i] > 0) r = i + d2[i], l = i
- d2[i] + 1;
```

```
    }
}
```

5.4 prefix function

```
void prefix_function(string &s,int pf[]){
    int n = s.size(); pf[0] = 0;
    for (int i=1; i<n; i++){
        int j = pf[i-1];
        while (j>0 && s[j] != s[i]) j = pf[j-1];
        pf[i] = j + (s[i] == s[j]);
    }
} // by NekoRolly
```

5.5 suffix array

```
enum Comparison { LESS, EQUAL, GREATER };

Comparison getComparison(int a, int b) {
    if (a < b) return Comparison::LESS;
    if (a > b) return Comparison::GREATER;
    return Comparison::EQUAL;
}

const int ALPH = 256;
struct SuffixArray {
    vector<int> suffixArray;
    vector<int> lcp;
    // lcp[i] = largest common preffix of sa[i] and sa[i + 1] (in sorted list)
    vector<int> pos;

    vector<int> sortCyclic(string &s) { // O(n log n)
        int n = s.size();
        vector<int> p(n), c(n), cnt(max(ALPH, n), 0);
        // p[] : sorted array with the starting index of substrings of length 2^k
        // c[] : equivalence class
        // counting sort
        for (int i = 0; i < n; i++) cnt[s[i]]++;
        for (int i = 1; i < ALPH; i++) cnt[i] += cnt[i - 1];
        for (int i = 0; i < n; i++) p[--cnt[s[i]]] = i;

        c[p[0]] = 0;
        int classes = 1;
        for (int i = 1; i < n; i++) {
            if (s[p[i]] != s[p[i - 1]]) classes++;
            c[p[i]] = classes - 1;
        }
        // radix sort
        vector<int> pNew(n), cNew(n);
        for (int k = 0; (1 << k) < n; k++) {
            // sort pNew by the second substring
            for (int i = 0; i < n; i++) {
                pNew[i] = p[i] - (1 << k);
                if (pNew[i] < 0) pNew[i] += n;
                cnt[i] = 0;
            }
            // counting sort in the first substring
            for (int i = 0; i < n; i++) cnt[c[pNew[i]]]++;
            for (int i = 1; i < classes; i++) cnt[i] += cnt[i - 1];
            for (int i = n - 1; i >= 0; i--) {
                p[--cnt[c[pNew[i]]]] = pNew[i];
            }
            cNew[p[0]] = 0;
            classes = 1;
            for (int i = 1; i < n; i++) {
                int x = p[i] + (1 << k);
                if (x >= n) x -= n;
                int y = p[i - 1] + (1 << k);
                if (y >= n) y -= n;
                pair<int, int> cur = {c[p[i]], c[x]};
                pair<int, int> prev = {c[p[i - 1]], c[y]};
                if (cur != prev) classes++;
                cNew[p[i]] = classes - 1;
            }
            c = cNew;
        }
        return p;
    }

    void buildLCP(string &s) { // O(n)
        int n = s.size();
        lcp = vector<int>(n - 1, 0);
        pos = vector<int>(n);
        for (int i = 0; i < n; i++) pos[suffixArray[i]] = i;
        int k = 0;
        for (int r = 0; r < n; r++) {
            if (pos[r] == 0) continue;
            int l = suffixArray[pos[r] - 1];
            while (s[l + k] == s[r + k]) k++;
            lcp[pos[r] - 1] = k;
            k = max(0, k - 1);
        }
    }

    // Suffix array: contain starting indexes of the all the suffixes (sorted)
    SuffixArray(string &s) { // O(n log n)
        char minChar = 'a' - 1;
        s += minChar;
        suffixArray = sortCyclic(s);
        suffixArray.erase(suffixArray.begin());
        s.pop_back();
        buildLCP(s);
    }

    // return the lcp of the suffixes s[l...n - 1] and s[r...n - 1]
    /*int getLCP(int l, int r) {
        assert(l != r);
        int minPos = min(pos[l], pos[r]);
        int maxPos = max(pos[l], pos[r]);
        return min(lcp[minPos], ..., lcp[maxPos - 1])
    }*/

    /*Comparison compare(int l1, int r1, int l2, int r2) {
        // compare two substrings [l1, r1], [l2, r2].
        int curLCP = (l1 == l2) ? suffixArray.size() : getLCP(l1, l2);
        if (r1 < l1 + curLCP) {
            if (r2 < l2 + curLCP) {
                int sz1 = r1 - l1 + 1;
                int sz2 = r2 - l2 + 1;
                return getComparison(sz1, sz2);
            } else return Comparison::LESS;
        } else if (r2 < l2 + curLCP) return Comparison::GREATER;
        return getComparison(pos[l1], pos[l2]);
    }*/
};
```

5.6 z function

```
void z_function(string &s,int zf[]){
    int n = s.size();
    for (int i=1,l=0,r=0; i<n; i++){
        if (i < r) zf[i] = min(zf[i-l], r-i);
        while (i+zf[i] < n && s[i+zf[i]] == s[zf[i]]) zf[i]++;
        if (i+zf[i] > r) l = i, r = i+zf[i];
    }
} // by NekoRolly
```